LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	29
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Typical cabin pressurization system, Cabin pressure regulator safety v/v, outflow v/v etc.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Typical cabin pressurization system, Cabin pressure regulator safety v/v, outflow v/v etc. and its operation, Methods of sealing of cabin
5. Specific Objectives	17.2.2.1 Trainee will be able to understand the necessity of cabin pressurization and working principles of components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of system schematic diagrams and identification of regulator and outflow valve components
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Why is cabin pressurization necessary for high-altitude flight?
(b) What are the primary sources of bleed air for the cabin system?
(c) What happens to the cabin pressure if the outflow valve fails fully open?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Necessity of cabin pressurization • Typical cabin pressurization system layout • Cabin pressure regulator function • Safety valve operation principles • Outflow valve control mechanisms	L & Ds	PPT, IB	15 min
(b)	• Methods of sealing the cabin • Pressure differential limits • Regulator safety valve interlocks • Outflow valve modes of operation • System response to altitude changes	L & Ds	PPT, PDF	20 min
(c)	• Component failure scenarios • Emergency depressurization procedures • Maintenance access points • System testing protocols • Troubleshooting common faults	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the safety valve in preventing over-pressurization.
(b) Describe the sequence of events during an emergency depressurization.
(c) How does the outflow valve maintain the desired cabin altitude?

12. Home Assignment and Follow Up	Prepare a brief note comparing the operation of the cabin pressure regulator with the air conditioning system components covered in LP 30.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Typical air conditioning system, various components, types of systems,	Refer to the master syllabus reference list	LP 30

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min